

3(a)(i)	polymerase chain reaction / PCR ;	1
3(a)(ii)	<i>any one from:</i> so that, DNA is / fragments are, visible (on gel) / AW ; (only need) a small starting quantity of DNA (from embryo) / AW ;	1
3(b)	<i>any four from:</i> Hb^S allele has, change in base sequence / mutation , in <i>Mst</i>II restriction site ; <i>Mst</i>II can no longer cut the Hb^S allele ; - difference in number of restriction sites ; (Hb^A allele has 3 / Hb^S allele has 2) - difference in number of fragments ; (Hb^A allele has 4 / Hb^S allele has 3) <i>Mst</i>II produces different sized DNA fragments when incubated with Hb^A and Hb^S alleles ; - difference in size of, main / middle, fragment ; A approx. 1200 bases and 200 bases v approx. 1400 bases	4
3(c)	<i>any three from:</i> - DNA / phosphate groups, negatively-charged ; - moves to anode ; - due to electric field / when current applied ; - larger / longer, fragments move, more slowly / less far ; ora - ref. to gel, impedance / resistance ; - ref. to buffer ;	3

Question	Answer	Marks										
3(d)(i)	used as a comparison (to show correct position of Hb^A and Hb^S on gel) ;	1										
3(d)(ii)	<table><tr><td>embryo</td><td>genotype</td></tr><tr><td>1</td><td>Hb^A Hb^A</td></tr><tr><td>2</td><td>Hb^A Hb^S</td></tr><tr><td>3</td><td>Hb^S Hb^S</td></tr><tr><td>4</td><td>Hb^A Hb^S</td></tr></table> one mark for correctly identifying sample 3 as Hb^S Hb^S ; one mark for correctly identifying samples 2 <u>and</u> 4 as Hb^A Hb^S ;	embryo	genotype	1	Hb ^A Hb ^A	2	Hb^A Hb^S	3	Hb^S Hb^S	4	Hb^A Hb^S	2
embryo	genotype											
1	Hb ^A Hb ^A											
2	Hb^A Hb^S											
3	Hb^S Hb^S											
4	Hb^A Hb^S											
3(e)	<i>any three from:</i> <i>pros:</i> 1 can avoid having offspring with, serious / genetic, disease ; A named example 2 can avoid late abortions (if genetic disease discovered later in foetal development) ; 3 allows couples to have children who would otherwise choose not to (due to risk of genetic disease) ; <i>cons:</i> 4 viable embryo(s) discarded ; R abortion 5 <i>idea of</i> use of healthcare resources by couple that can conceive naturally ; 6 may conflict with religious beliefs ; 7 could lead to selection based on gender or specific traits (“designer babies”) ; 8 AVP ; e.g. genetic disease may not develop	3										